Experiment - 2

Aim:

Convert the ER diagram into a relational table schema. Write complete MySQL CREATE TABLE statements with Primary key, foreign key, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample data and demonstrate referential integrity violations.

Relational schema

Student
- Student_id PRIMARY KEY
- name
- Email (unique)
- Dept_id FOREIGN KEY

Department
- Dept_id PRIMARY KEY
- Dept_name UNIQUE

Course
- Course_id PRIMARY KEY
- Course_name
- credits

Enrollment
- student_id PRIMARY KEY
- course_id
- Enroll_data

Source code (MySQL)

```
CREATE TABLE Department(
        Dept_id INT PRIMARY KEY,
        Dept_name VARCHAR(50) NOT NULL UNIQUE

);
CREATE TABLE students (
        student_id INT PRIMARY KEY,
        name VARCHAR (50) NOT NULL,
        email VARCHAR (100) Unique,
        Dept_id INT
        FOREIGN KEY (Dept_id)
                REFERENCES Department (Dept_id)
                ON DELETE SET NULL

);
CREATE TABLE course(
        course_id INT PRIMARY KEY,
        course_name VARCHAR (100) NOT NULL,
        credits INT NOT NULL

);
CREATE TABLE Enrollment (
        student_id INT,
        course_id INT,
        enroll_date DATE NOT NULL,
        PRIMARY KEY (student_id, course_id),

        FOREIGN KEY (student_id)
                REFERENCES student (student_id)
                ON DELETE CASCADE,
```

```
    FOREIGN KEY (course_id)
        REFERENCES course (course_id)
        ON DELETE CASCADE
);
```

Insert sample Data

```
INSERT INTO department VALUES
(1, 'CSE'),
(2, 'ECE');
INSERT INTO students VALUES
(101, 'Rohit', 'rohit@gmail.com', 1),
(102, 'Aditya', 'adityalo@gmail.com', 2);
INSERT INTO course VALUES
(201, 'DBMS', 4),
(202, 'Python', 3);
INSERT INTO ENROLLMENT VALUES
(101, 201, '2026-08-19'), (101, 202, '2026-08-19'),
(102, 201, '2026-07-20');
```

Output

```
SELECT FROM students;
```

student_id	name	email	dept_id
1	Rohit	rohit@gmail.com	1
2	Aditya	@adityalo@gmail.com	2

SELECT FROM Enrollment;

student_id	course_id	enroll_data
101	201	2026-08-19
101	202	2026-08-19
102	201	2026-07-20

SELECT FROM Department;

Dept_id	Dept_name
1	CSE
2	ECE